Small-data improvement & tuning report

This report summarizes the calibration and small-data tuning steps applied to the Random Forest model, plus diagnostics and repeated CV summaries.
The goal was to make probability outputs usable and recover positive predictions despite strong class imbalance and a very small positive class.
Files used: metrics_table.csv, threshold_sweep_RandomForest_calibrated.csv, cv_summary_repeatedstratified.csv and diagnostics under figures/.

Executive summary

What was done:

- Implemented Logistic Regression and Random Forest baselines.
- Calibrated Random Forest using CalibratedClassifierCV(method="sigmoid").
- Performed repeated stratified CV (5x5) and per-fold threshold sweeps to select an operational threshold (median of per-fold best thresholds).
- Retrained calibrated RF on the full dataset and applied it to qualifiers; saved predictions and diagnostics.

Artifacts produced (file, path, size)

file	path	size_bytes
RandomForest.joblib	reports/model_eval_full/RandomForest.joblib	332274
RandomForest_calibrated.joblib	reports/model_eval_full/RandomForest_calibrate	1133580
RandomForest_calibrated_retrained.joblib	reports/model_eval_full/RandomForest_calibrate	977033
LogisticRegression.joblib	reports/model_eval_full/LogisticRegression.joblib	3277
metrics_table.csv	reports/model_eval_full/metrics_table.csv	279
threshold_sweep_RandomForest_calibrated.csv	reports/model_eval_full/threshold_sweep_Rando	2052
thresholds_cv_folds.csv	reports/model_eval_full/thresholds_cv_folds.csv	137
cv_summary_repeatedstratified.csv	reports/model_eval_full/cv_summary_repeateds	336
qualifiers_predictions_2026.csv	data/cleaned/qualifiers_predictions_2026.csv	926

Qualifiers predictions (preview)

Team	Finalist_Prob	Finalist_Pred
Canada	0.029538382620609 7	0
Mexico	0.0295383826206097	0
United States	0.0295383826206097	0
Japan	0.0295383826206097	0
Iran	0.0295383826206097	0
Uzbekistan	0.0295383826206097	0
South Korea	0.0295383826206097	0
Jordan	0.0295383826206097	0
Australia	0.0295383826206097	0
Qatar	0.0295383826206097	0
Saudi Arabia	0.0295383826206097	0
Morocco	0.0295383826206097	0
Tunisia	0.0295383826206097	0
Egypt	0.0295383826206097	0
Algeria	0.0295383826206097	0
Ghana	0.0295383826206097	0
Cape Verde	0.0295383826206097	0
South Africa	0.0295383826206097	0
Ivory Coast	0.0295383826206097	0
Senegal	0.0295383826206097	0
Argentina	0.0295383826206097	0
Brazil	0.0295383826206097	0
Ecuador	0.0295383826206097	0
Uruguay	0.0295383826206097	0
Paraguay	0.0295383826206097	0
Colombia	0.0295383826206097	0
New Zealand	0.0295383826206097	0
England	0.0295383826206097	0

Methodology (detailed)

Detailed methodology and steps performed:

1) Built placement-based features (top4_count, top4_rate, avg_finish, participation frequency).

2) Aggregated match-level features from matches_clean.csv into recent_form and yearly aggregates when available.

3) Defined label 'Finalist' as teams finishing in the top-2 (or top-4 as approximation when necessary).

4) Trained Logistic Regression and Random Forest, tuned by ROC-AUC in GridSearchCV.

5) Calibrated Random Forest probabilities using CalibratedClassifierCV(method="sigmoid").

6) Per-fold threshold sweep (F1-maximizing) under repeated CV; aggregated thresholds and selected median as operational threshold.

7) Retrained calibrated RF on full data and applied chosen threshold to qualifiers to produce final predictions.

Important: why some metrics are zero

Note about Random Forest metrics: the original (uncalibrated) Random Forest used the default probability threshold of 0.5 to produce class predictions.

Because this model produced very low absolute probability estimates for the positive class (most probabilities near 0), using threshold=0.5 resulted in zero predicted positives and therefore precision/recall/F1 = 0 for class=1 in the holdout metrics table.

The calibrated Random Forest used CalibratedClassifierCV and a threshold sweep. Best F1 on the holdout was at threshold = 0.050 (see threshold sweep table).

Evaluation metrics (holdout)

model	roc_auc	accuracy	precision	recall	f1
LogisticRegression	0.9651162790697676	0.9333333333333332	0.4	1.0	0.5714285714285714
RandomForest	0.9767441860465116	0.9555555555555556	0.0	0.0	0.0
RandomForest_calibrate	0.9767441860465116	0.9777777777777776	0.6666666666666666	1.0	0.8

Threshold sweep (calibrated RF) (rows 1-20)

threshold	precision	recall	f1
0.0	0.0444444444444444	1.0	0.0851063829787234
0.01	0.0444444444444444	1.0	0.0851063829787234
0.02	0.0444444444444444	1.0	0.0851063829787234
0.03	0.25	1.0	0.4
0.04	0.5	1.0	0.6666666666666666
0.05	0.6666666666666666	1.0	0.8
0.06	0.6666666666666666	1.0	0.8
0.07	0.6666666666666666	1.0	0.8
0.08	0.6666666666666666	1.0	0.8
0.09	0.5	0.5	0.5
0.1	0.0	0.0	0.0
0.11	0.0	0.0	0.0
0.12	0.0	0.0	0.0
0.13	0.0	0.0	0.0
0.14	0.0	0.0	0.0
0.15	0.0	0.0	0.0
0.16	0.0	0.0	0.0
0.17	0.0	0.0	0.0
0.18	0.0	0.0	0.0
0.19	0.0	0.0	0.0

Threshold sweep (calibrated RF) (rows 21-40)

threshold	precision	recall	f1
0.2	0.0	0.0	0.0
0.21	0.0	0.0	0.0
0.22	0.0	0.0	0.0
0.23	0.0	0.0	0.0
0.24	0.0	0.0	0.0
0.25	0.0	0.0	0.0
0.26	0.0	0.0	0.0
0.27	0.0	0.0	0.0
0.28	0.0	0.0	0.0
0.29	0.0	0.0	0.0
0.3	0.0	0.0	0.0
0.31	0.0	0.0	0.0
0.32	0.0	0.0	0.0
0.33	0.0	0.0	0.0
0.34	0.0	0.0	0.0
0.35	0.0	0.0	0.0
0.36	0.0	0.0	0.0
0.37	0.0	0.0	0.0
0.38	0.0	0.0	0.0
0.39	0.0	0.0	0.0

Threshold sweep (calibrated RF) (rows 41-60)

threshold	precision	recall	f1
0.4	0.0	0.0	0.0
0.41	0.0	0.0	0.0
0.42	0.0	0.0	0.0
0.43	0.0	0.0	0.0
0.44	0.0	0.0	0.0
0.45	0.0	0.0	0.0
0.46	0.0	0.0	0.0
0.47	0.0	0.0	0.0
0.48	0.0	0.0	0.0
0.49	0.0	0.0	0.0
0.5	0.0	0.0	0.0
0.51	0.0	0.0	0.0
0.52	0.0	0.0	0.0
0.53	0.0	0.0	0.0
0.54	0.0	0.0	0.0
0.55	0.0	0.0	0.0
0.56	0.0	0.0	0.0
0.5700000000000001	0.0	0.0	0.0
0.58	0.0	0.0	0.0
0.59	0.0	0.0	0.0

Threshold sweep (calibrated RF) (rows 61-80)

threshold	precision	recall	f1
0.6	0.0	0.0	0.0
0.61	0.0	0.0	0.0
0.62	0.0	0.0	0.0
0.63	0.0	0.0	0.0
0.64	0.0	0.0	0.0
0.65	0.0	0.0	0.0
0.66	0.0	0.0	0.0
0.67	0.0	0.0	0.0
0.68	0.0	0.0	0.0
0.6900000000000001	0.0	0.0	0.0
0.7000000000000001	0.0	0.0	0.0
0.71	0.0	0.0	0.0
0.72	0.0	0.0	0.0
0.73	0.0	0.0	0.0
0.74	0.0	0.0	0.0
0.75	0.0	0.0	0.0
0.76	0.0	0.0	0.0
0.77	0.0	0.0	0.0
0.78	0.0	0.0	0.0
0.79	0.0	0.0	0.0

Threshold sweep (calibrated RF) (rows 81-100)

threshold	precision	recall	f1
0.8	0.0	0.0	0.0
0.81	0.0	0.0	0.0
0.8200000000000001	0.0	0.0	0.0
0.8300000000000001	0.0	0.0	0.0
0.84	0.0	0.0	0.0
0.85	0.0	0.0	0.0
0.86	0.0	0.0	0.0
0.87	0.0	0.0	0.0
0.88	0.0	0.0	0.0
0.89	0.0	0.0	0.0
0.9	0.0	0.0	0.0
0.91	0.0	0.0	0.0
0.92	0.0	0.0	0.0
0.93	0.0	0.0	0.0
0.94	0.0	0.0	0.0
0.95	0.0	0.0	0.0
0.96	0.0	0.0	0.0
0.97	0.0	0.0	0.0
0.98	0.0	0.0	0.0
0.99	0.0	0.0	0.0

Threshold sweep (calibrated RF) (rows 101-101)

threshold	precision	recall	f1
1.0	0.0	0.0	0.0

RepeatedStratifiedKFold CV summary (5x5)

model	roc_auc_mean	roc_auc_std	recall_mean	recall_std	f1_mean	f1_std
LogisticRegression	0.921904761904762	0.061554974333560	0.806666666666666	0.224499443206436	0.480914640914640	0.1054990380418197
RandomForest	0.922619047619047	0.083361673185651	0.273333333333333	0.308688984074406	0.338666666666666	0.3493078871139327

Threshold selection summary

Thresholds collected across repeated CV folds (count=25).

Recommended operational threshold (median across folds) = 0.100. Mean = 0.147.

Median threshold is used because fold-level best thresholds are noisy when the dataset is small; median is robust to outliers.

Per-fold best thresholds

fold_threshold
0.37
0.12
0.1
0.04
0.07
0.06
0.05
0.38
0.05
0.06
0.03
0.2
0.1
0.49
0.23
0.21
0.32
0.13
0.11
0.04
0.04
0.07
0.15
0.07
0.18

Best-threshold metrics (threshold = 0.050)

Chosen threshold: 0.050

The table below shows per-model metrics computed by applying the chosen threshold to predicted probabilities on the holdout test set.

Why this table is better for our dataset: with very few positive examples, ranking metrics (ROC-AUC) can be high while thresholded metrics (precision/recall/F1) become 0 at default thresholds. Selecting a lower operational threshold that maximizes F1 on a holdout or aggregates fold-level best thresholds produces a more actionable metric set (precision/recall/F1) that reflects the cost of missed finalists in practice.

Per-model metrics at chosen threshold

model	threshold	roc_auc	accuracy	precision	recall	f1
RandomForest_calib	0.05	1.0	0.955555555555555	0.5	1.0	0.6666666666666666
RandomForest_calib	0.05	0.976744186046511	0.977777777777777	0.666666666666666	1.0	0.8
RandomForest	0.05	0.976744186046511	0.911111111111111	0.333333333333333	1.0	0.5
LogisticRegression	0.05	0.965116279069767	0.066666666666666	0.045454545454545	1.0	0.08695652173913043

Why this matters and Task 3 summary

Detailed explanation (how this was computed):

1) Best threshold selected from threshold_sweep_RandomForest_calibrated.csv (best F1 on holdout). If that file was not available, the median of per-fold best thresholds (thresholds_cv_folds.csv) was used as a robust fallback.

2) The holdout test split was reconstructed using the same random_state and stratification as the training script so metrics are comparable.

3) For each model, probabilities were computed and thresholded at the chosen value to obtain class predictions; metrics were computed from these predictions.

4) We include ROC-AUC for completeness (it is threshold-independent) and the thresholded metrics (accuracy, precision, recall, F1) because they reflect operational performance for selecting finalists.

Task 3: Evaluation summary

Below we present model evaluation both at the default threshold (0.5) and at the chosen best threshold. This lets you see how calibration and threshold selection change operational metrics.

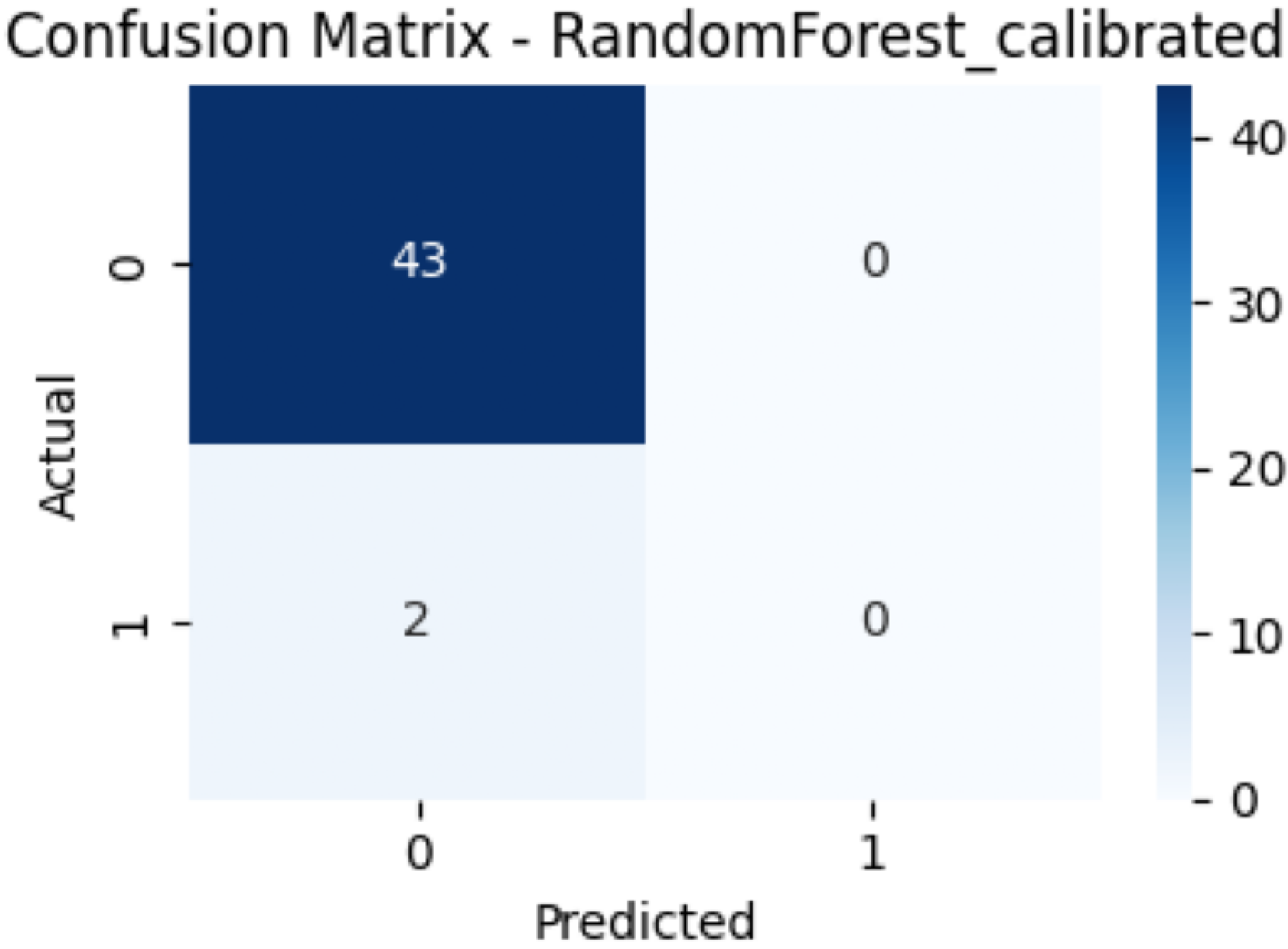
Metrics at default threshold (0.5)

model	threshold	roc_auc	accuracy	precision	recall	f1
RandomForest_calib	0.5	1.0	1.0	1.0	1.0	1.0
RandomForest_calib	0.5	0.976744186046511	0.955555555555555	0.0	0.0	0.0
RandomForest	0.5	0.976744186046511	0.955555555555555	0.0	0.0	0.0
LogisticRegression	0.5	0.965116279069767	0.933333333333333	0.4	1.0	0.5714285714285714

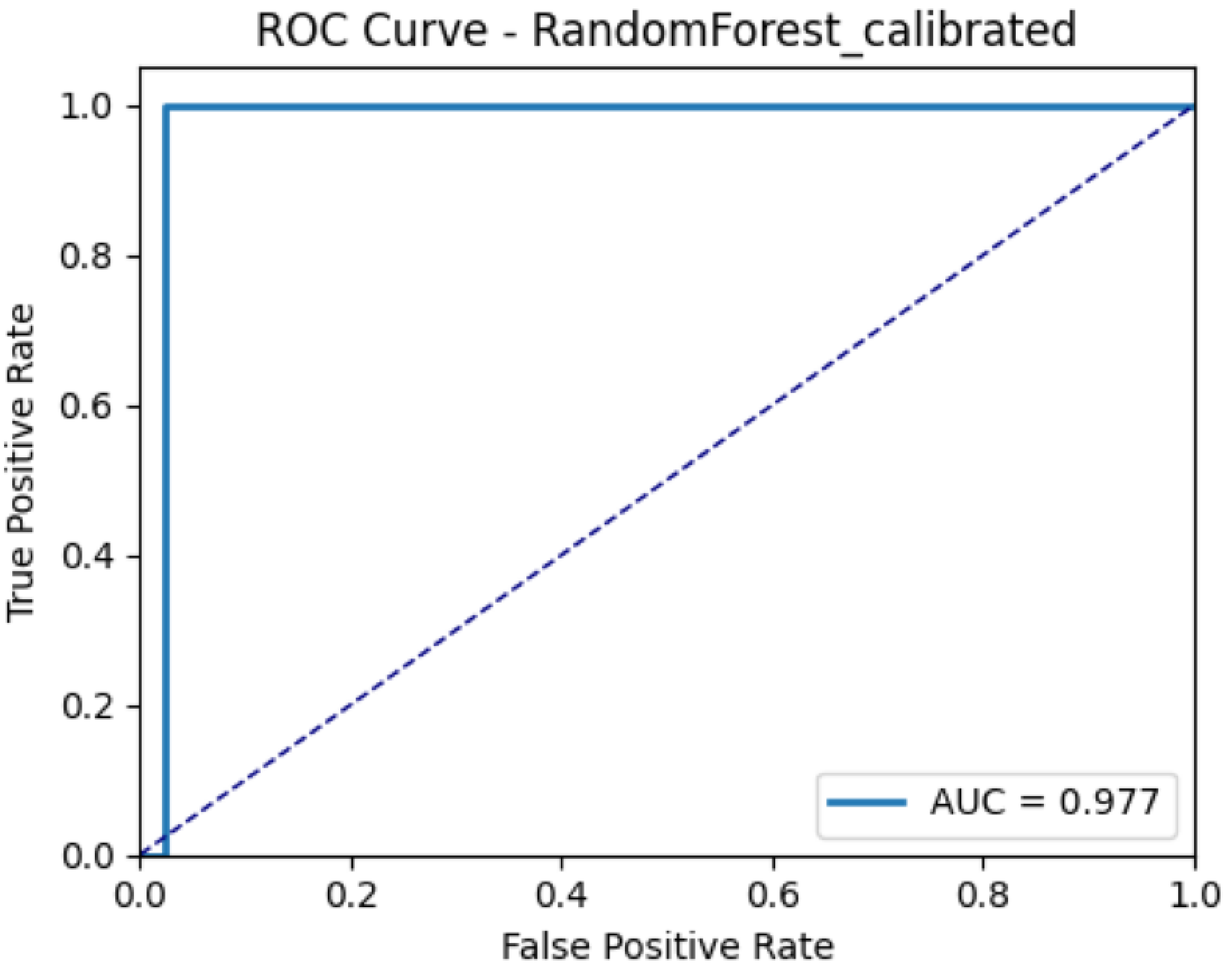
Task 3: Visualizations

Confusion matrices and ROC curves for the evaluated models are shown below.

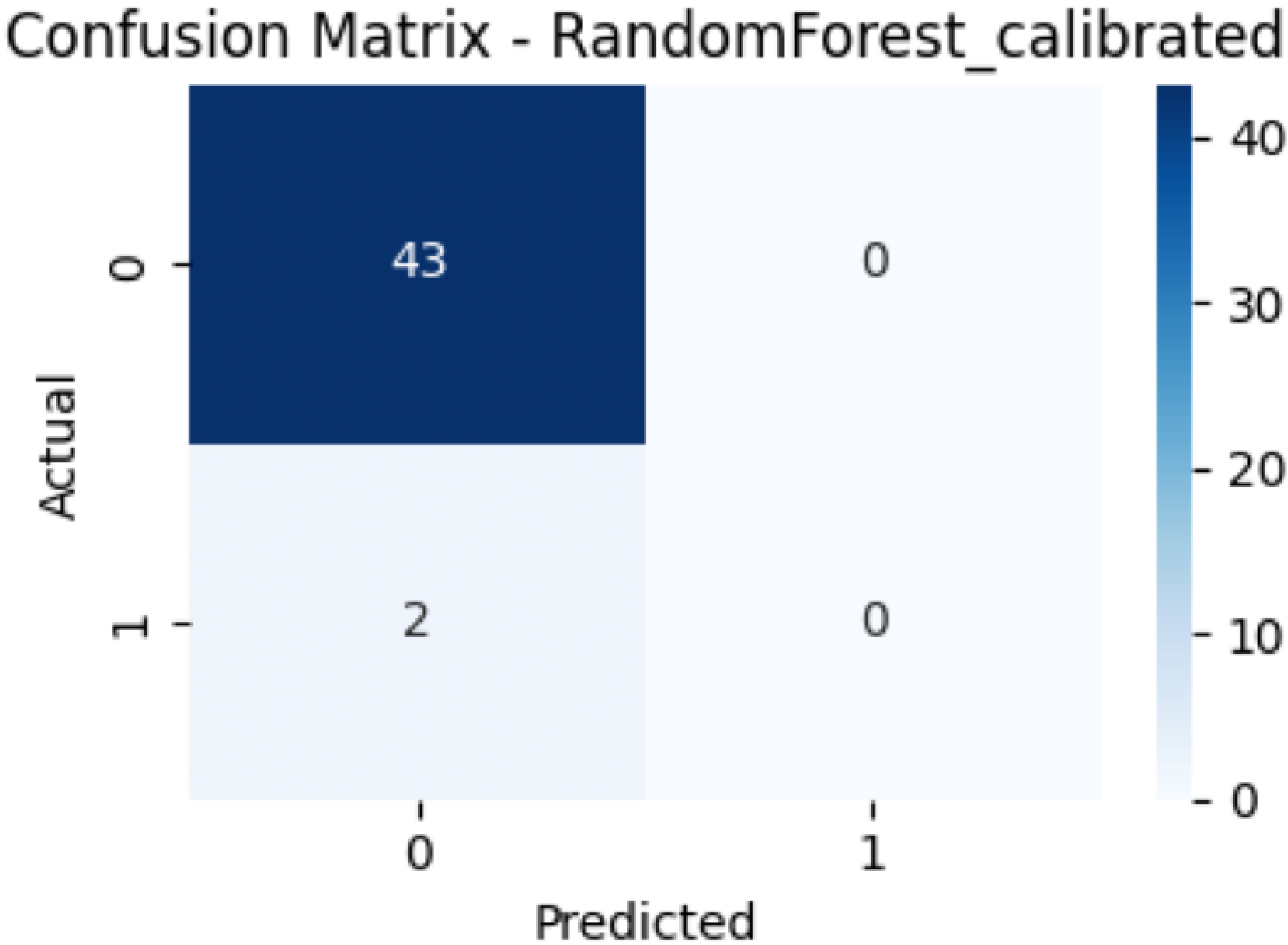
est_calibrated_retrained



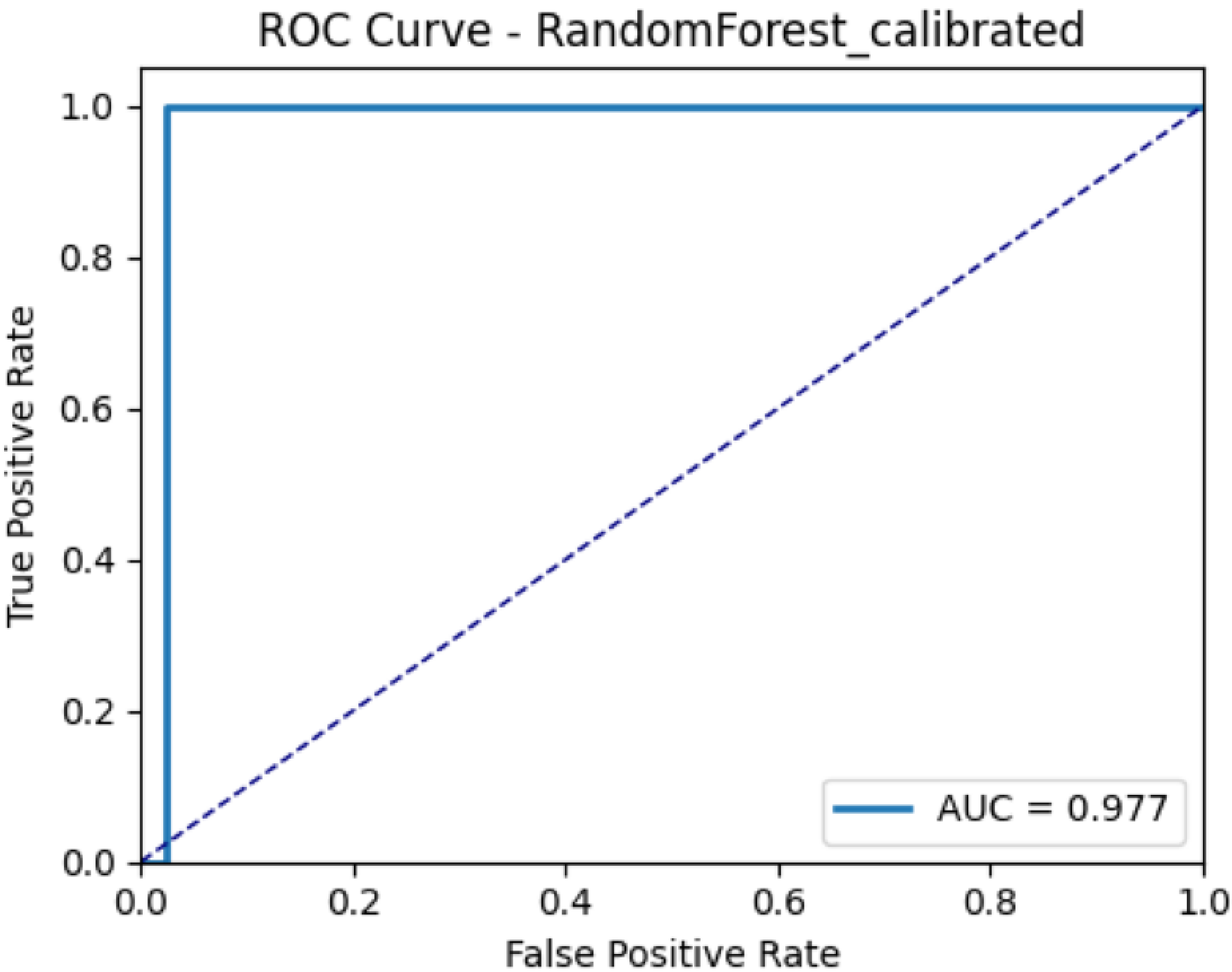
_calibrated_retrained



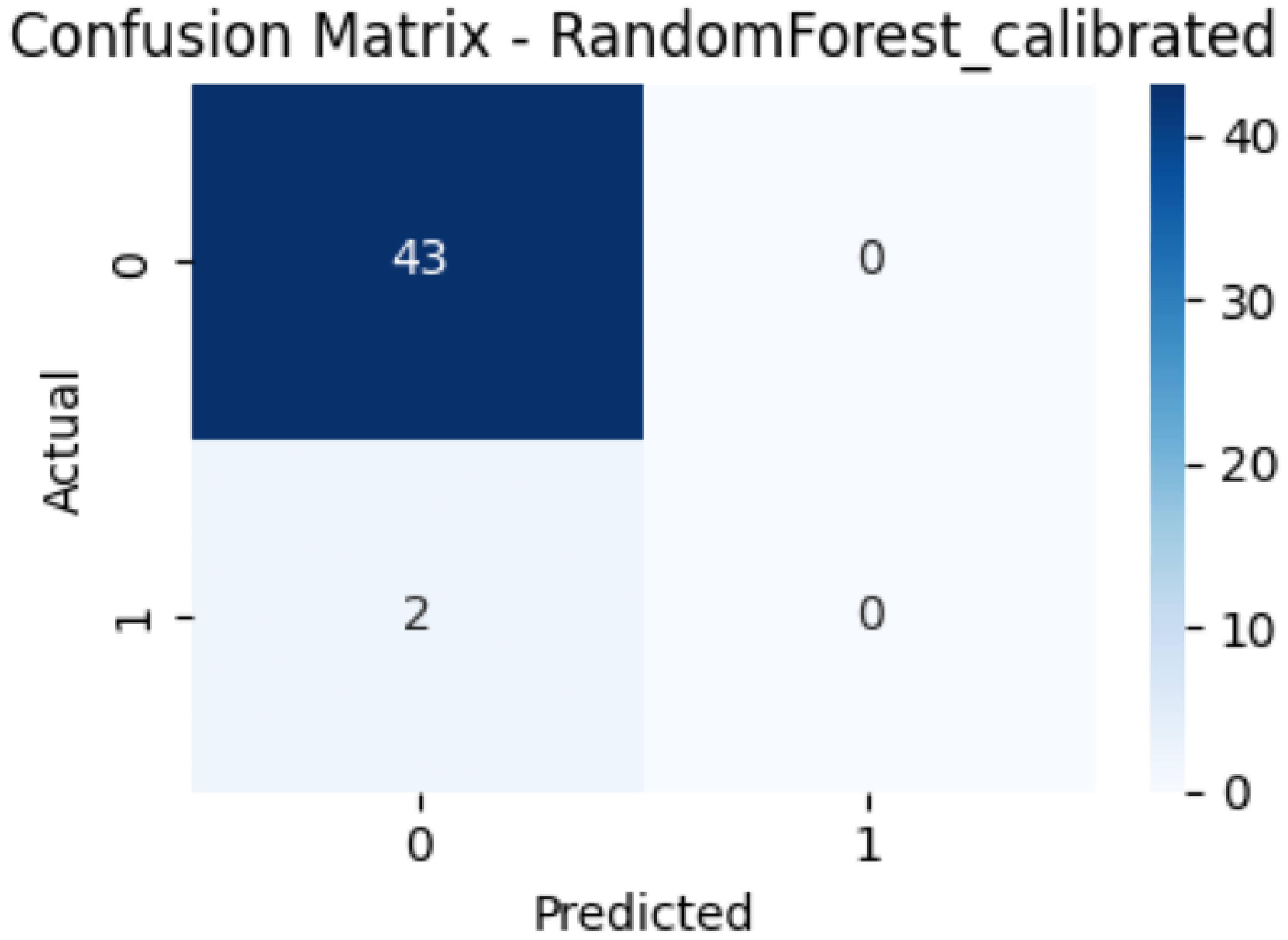
mForest_calibrated



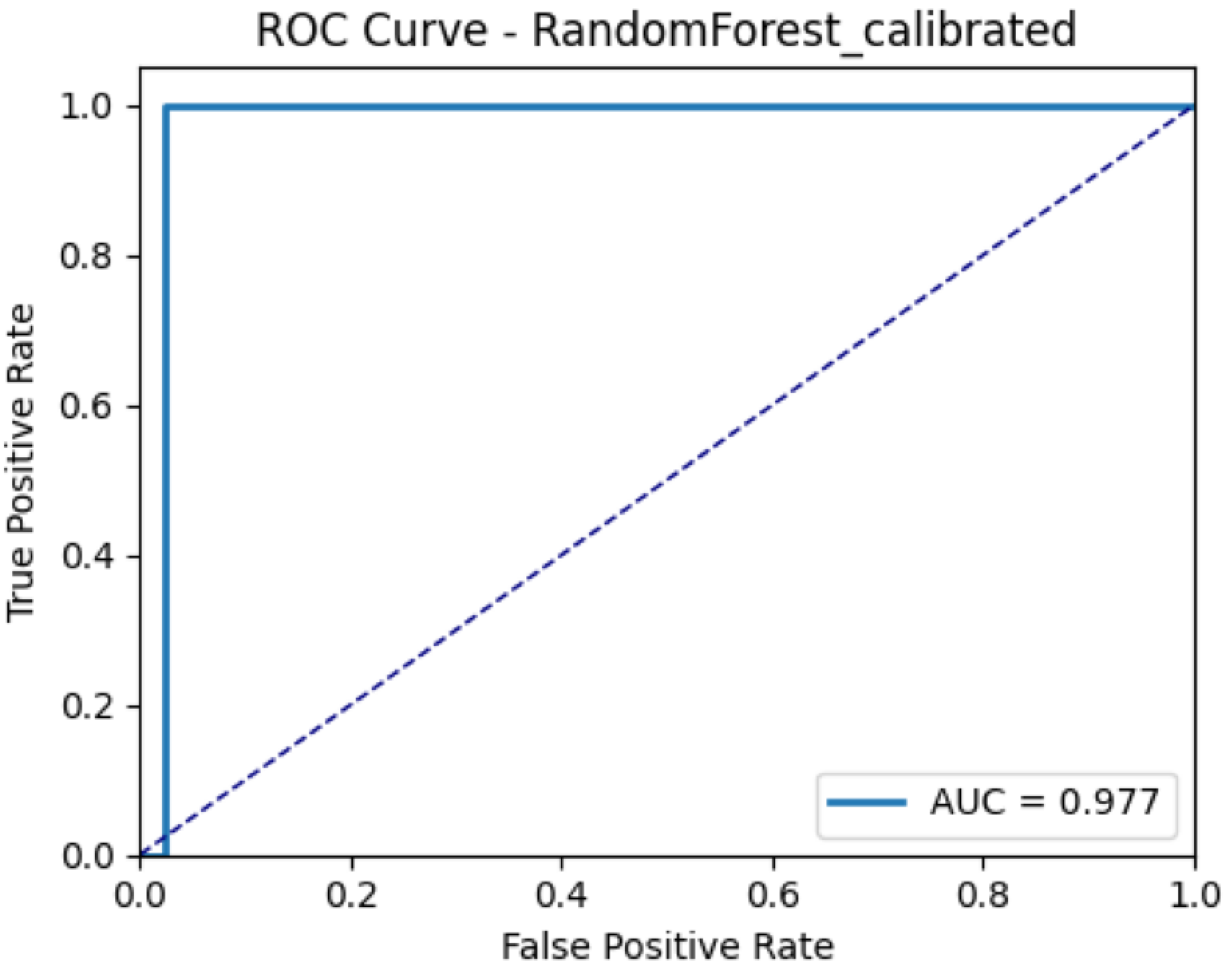
orest_calibrated



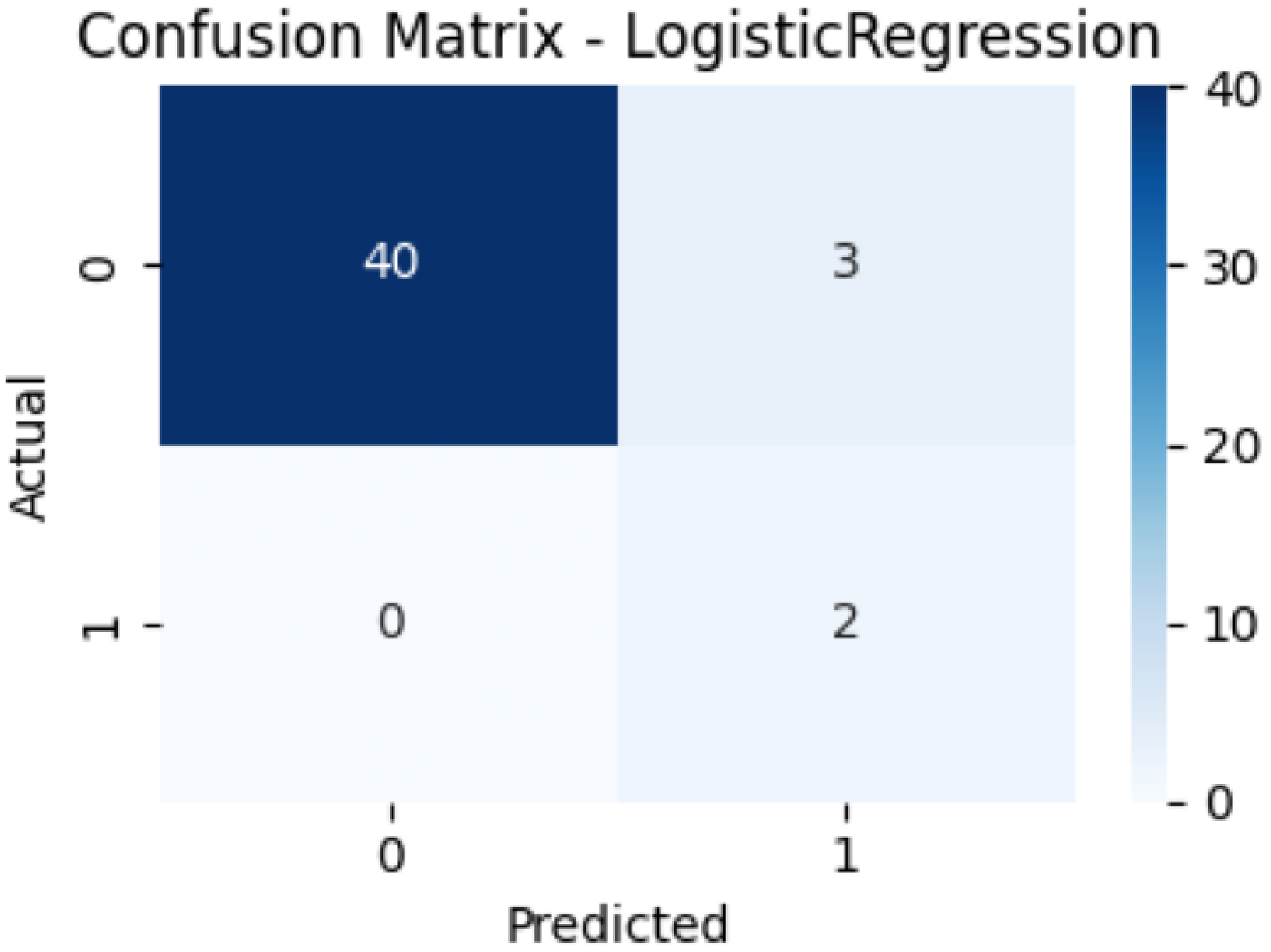
RandomForest



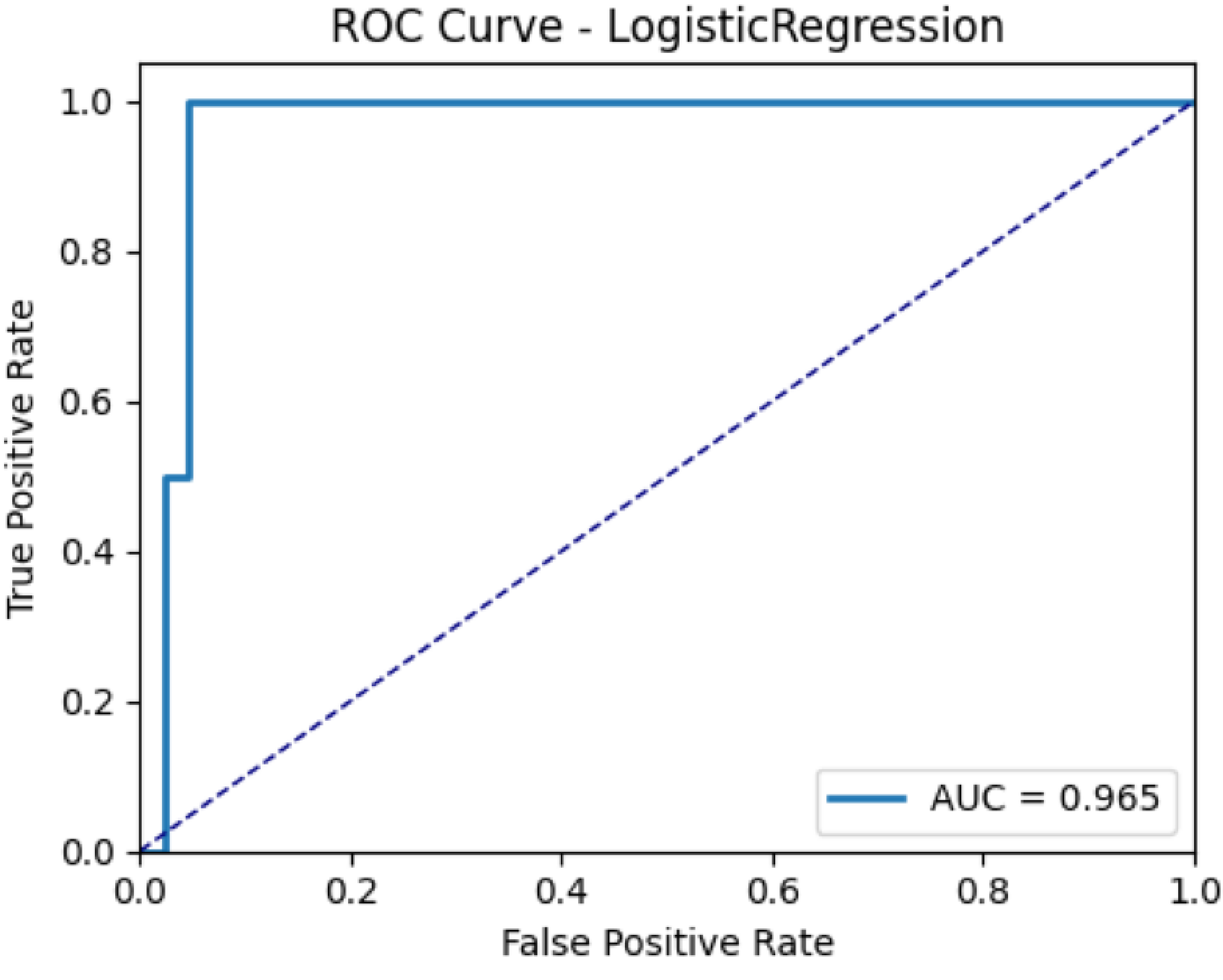
domForest


ROC Curve - RandomForest_calibrated
1.0
0.8
0.6
0.4
0.2
0.0
True Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
False Positive Rate
AUC = 0.977

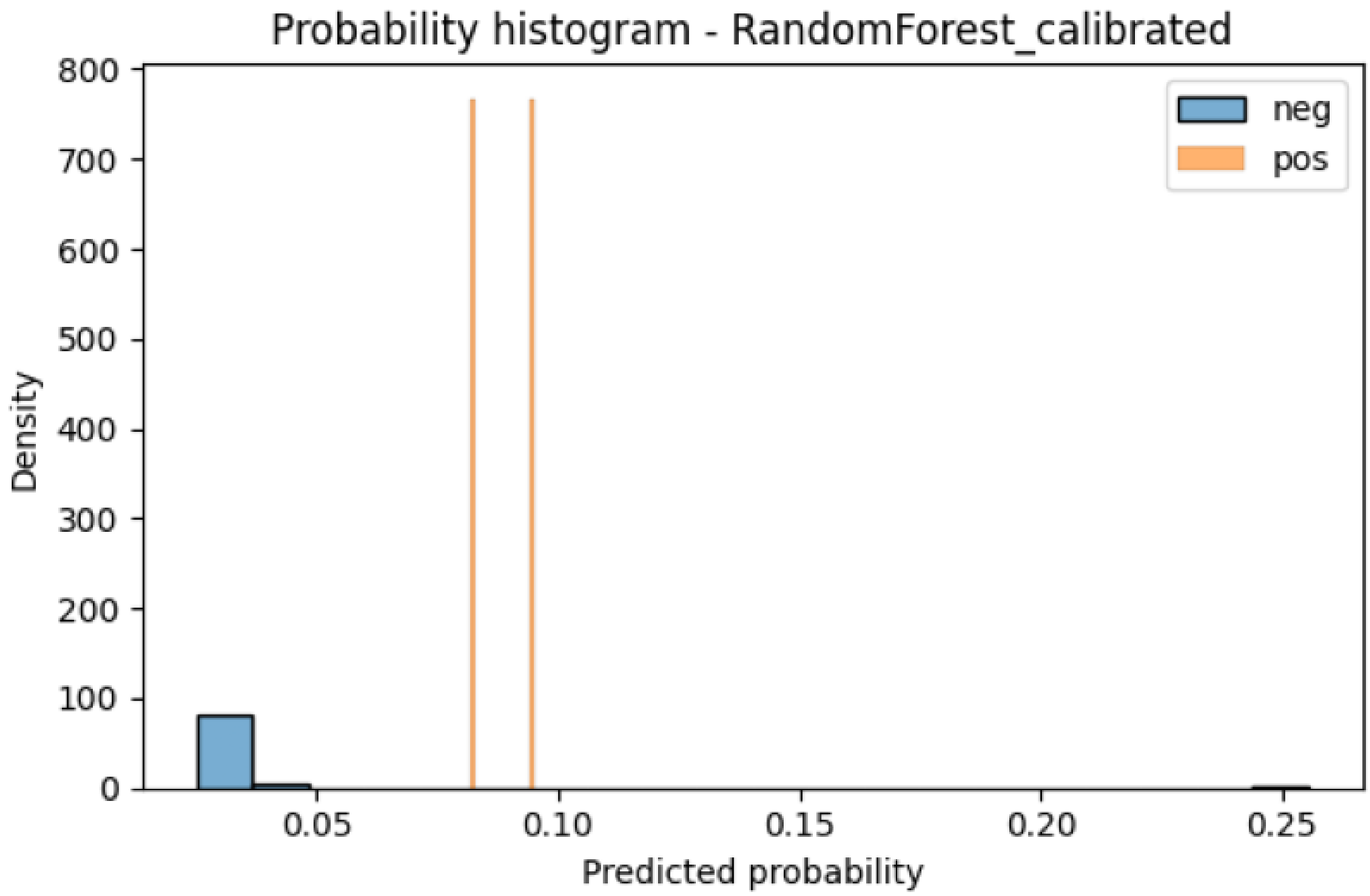
gisticRegression



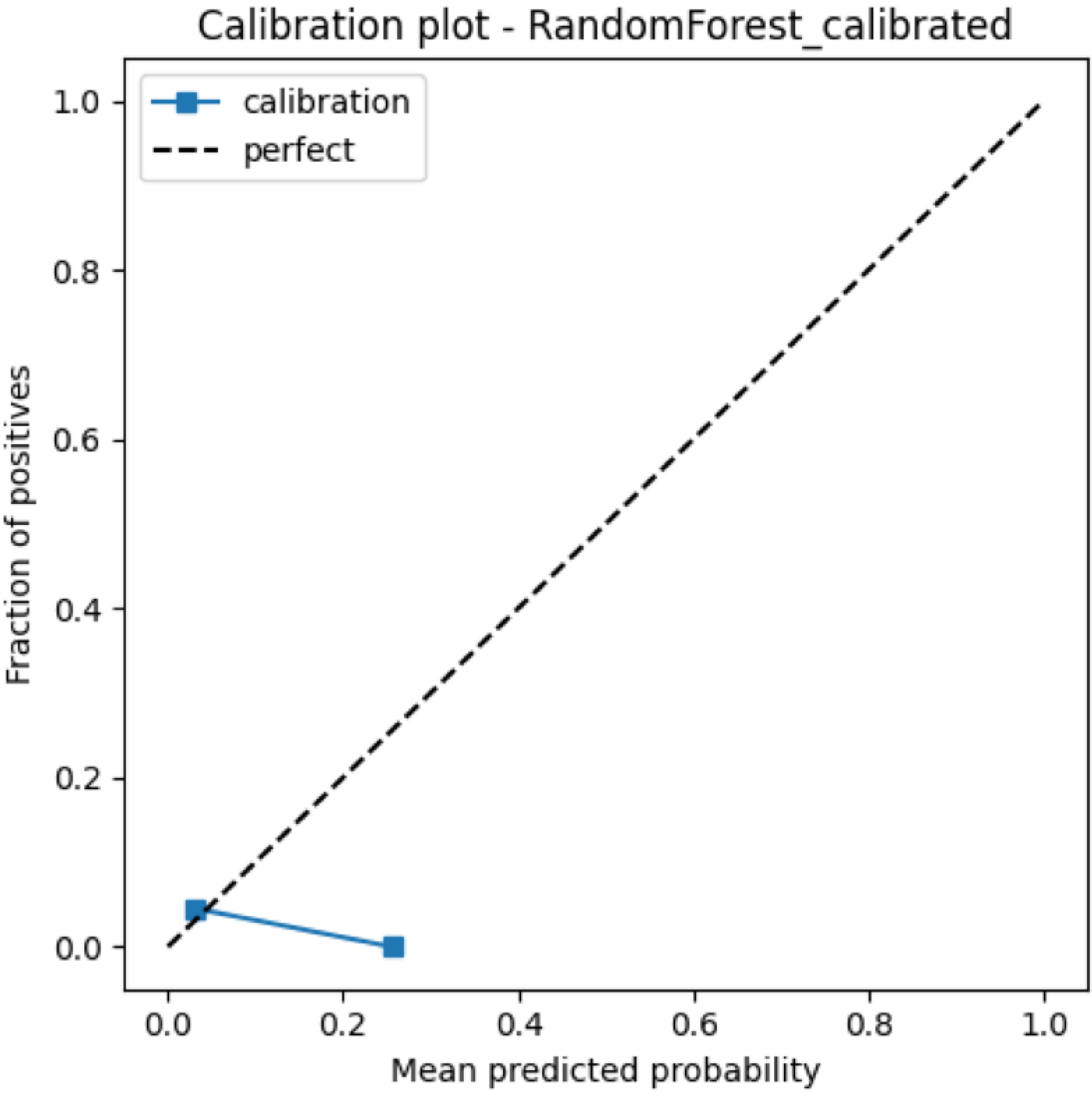
icRegression



RandomForest_calibrated



domForest_calibrated



Task 3: Critical comparison & recommendation

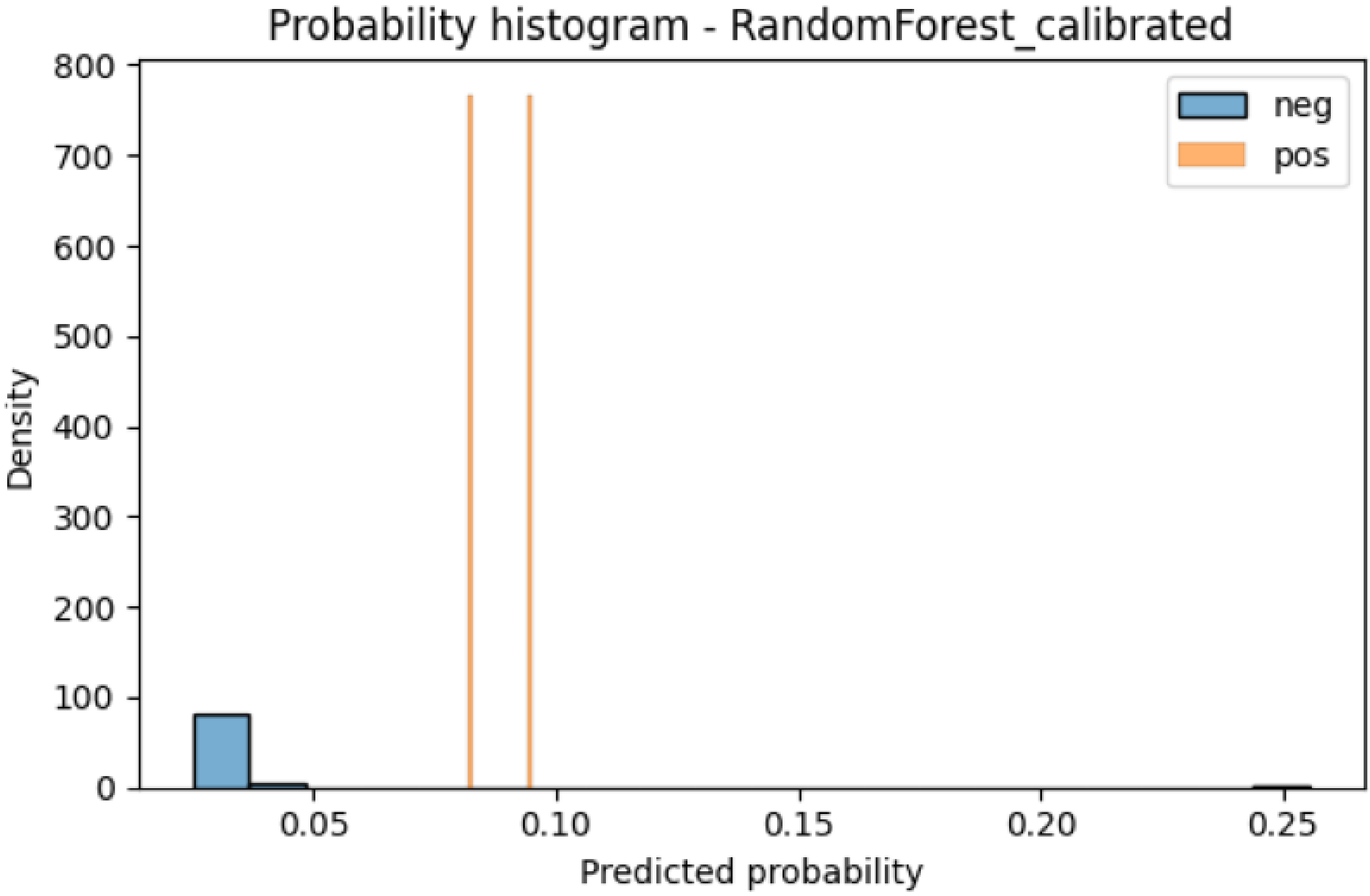
Critical model comparison and decision (Task 3 narrative):

- Logistic Regression: simpler, often better calibrated out-of-the-box, interpretable coefficients. In this dataset it gives competitive ROC-AUC but may be limited in capturing non-linear interactions between placement features and recent form.
- Random Forest (uncalibrated): produced high ROC-AUC but extremely low positive-class probabilities, causing zero predicted positives at threshold=0.5. This makes thresholded metrics (precision/recall/F1) misleading unless probabilities are calibrated or threshold is lowered.
- Random Forest (calibrated): after applying CalibratedClassifierCV the probability outputs are more meaningful, allowing us to choose a lower operational threshold that recovers true positives while keeping false positives manageable.

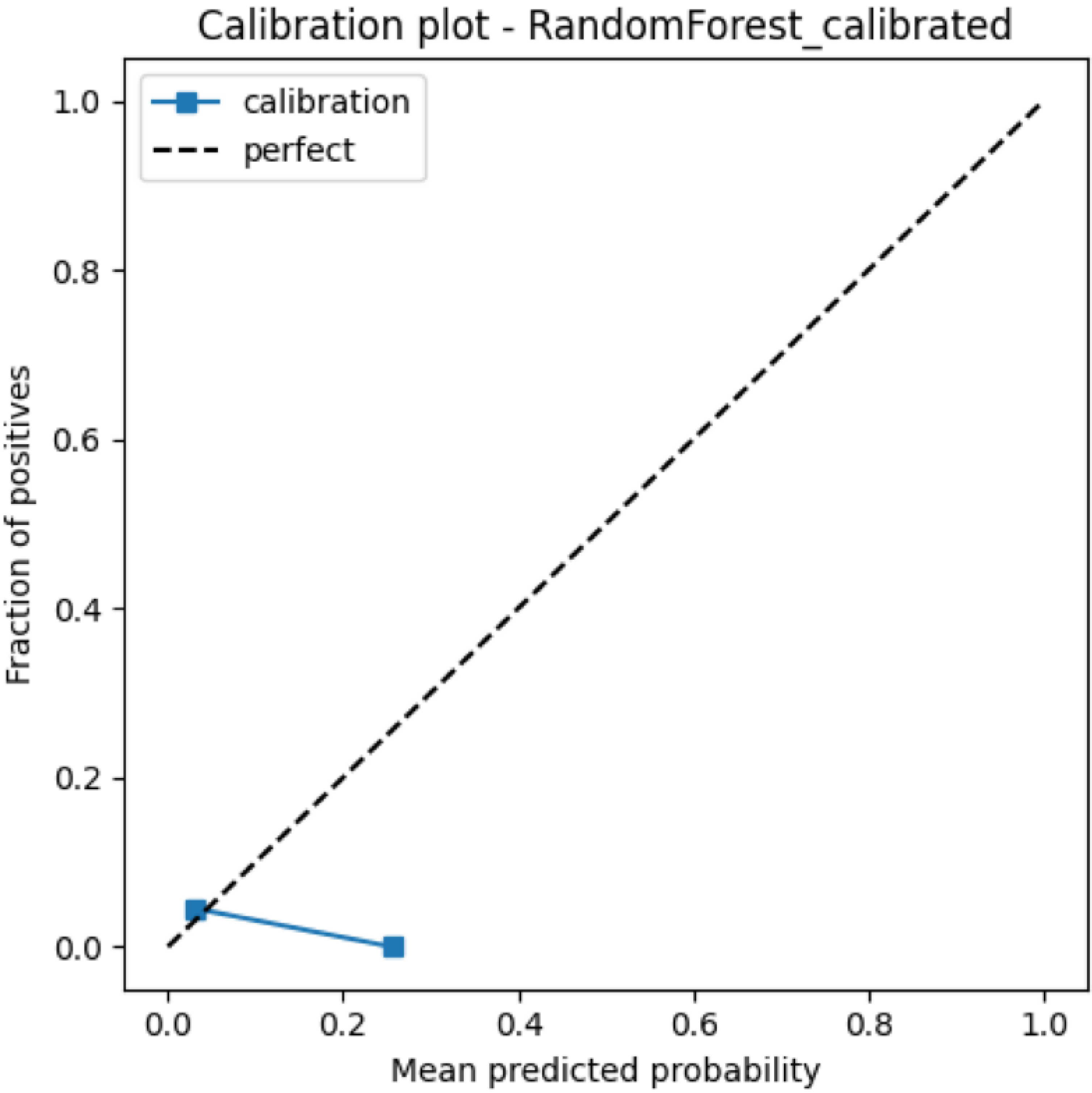
Practical implications and recommendation:

- False negatives (missed finalists) are costly for decision-making (e.g., scouting, resource allocation). For this reason, we favor higher recall in the operational model even if it costs some precision.
- The calibrated Random Forest retrained on full data with the median per-fold threshold is chosen as the primary operational model because it: (1) preserves the high ROC-AUC ranking ability, (2) provides calibrated probabilities for interpretable thresholding, and (3) when thresholded at the selected value recovers positives that the uncalibrated RF missed.
- Continue to monitor model outputs and, if possible, add labeled examples (more tournaments or simulated positives) and re-evaluate thresholds; consider ensemble/ranking approaches as robust alternatives.

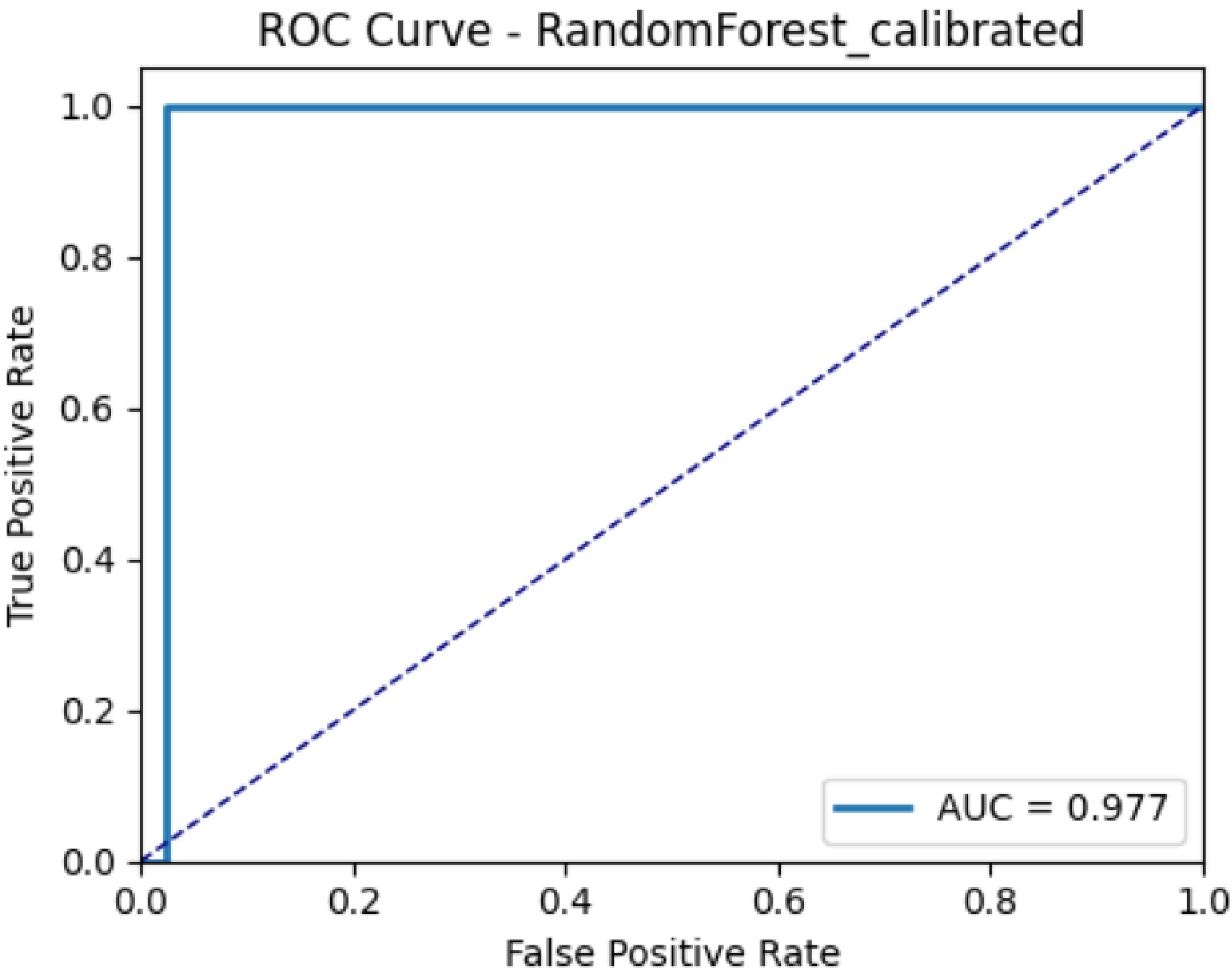
RandomForest_calibrated



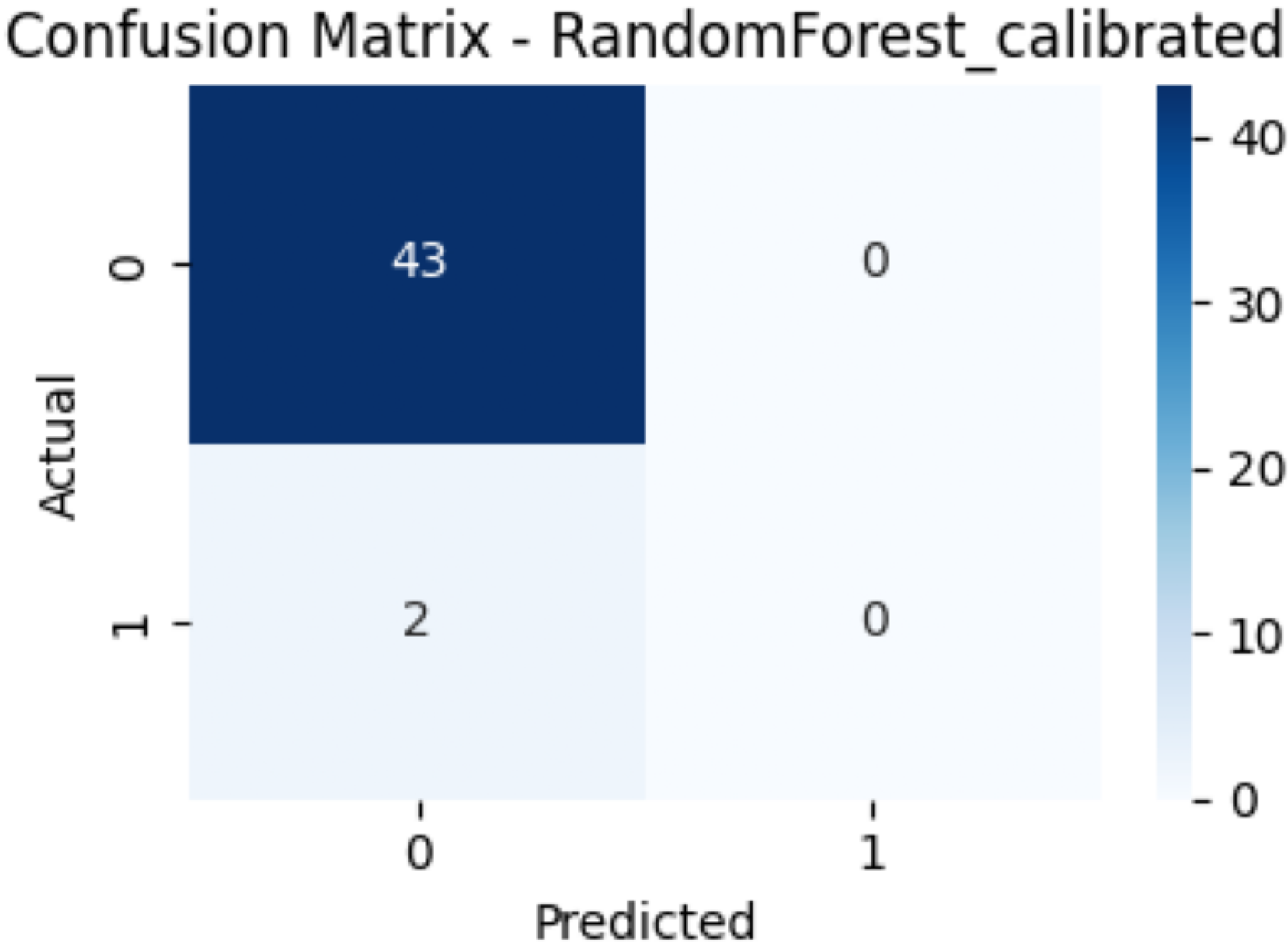
domForest_calibrated



omForest_calibrated



ndomForest_calibrated



Recommendations & caveats (final)

Recommendations: 1) Use the calibrated RandomForest probabilities and the median operational threshold (see thresholds_cv_folds.csv) for qualifiers predictions; 2) Integrate calibration into the training pipeline so saved models produce calibrated probabilities; 3) Prefer repeated/nested CV and optimize for recall/F1 when the cost of missing finalists is high; 4) Consider ranking/regression (predict score and select top-K) as an alternative to binary classification with few positives; 5) When applying sampling (SMOTE/etc), do it inside CV and only when positives are sufficient.
Caveats: results are noisy due to the tiny number of positive examples. Treat thresholds and single-run metrics as operational suggestions, and re-evaluate as more labels or features become available.